Prof. Dr.-Ing. Jan-Niklas Voigt-Antons

Professor of Computer Science (Immersive Media) · Hamm-Lippstadt University of Applied Sciences
Director, Immersive Reality Lab · Guest Researcher, Technische Universität Berlin

jan-niklas@voigt-antons.de · voigt-antons.de · ORCID 0000-0002-2786-9262 · Hamm · Lippstadt · Berlin, Germany

€4.609 M	**234**	**2,995**	**ITU-T P.812**
third-party funding	publications	citations, h-index 29	co-developer

I study how virtual and augmented reality systems can be built so they actually work for people. My work pairs EEG, eye tracking and physiological sensing with XR deployments in hospitals, public space and safety-critical training — measuring what questionnaires cannot capture.

APPOINTMENTS

since 2024	**Co-Head, Computer Visualistics and Design** degree programme (with a co-lead), HSHL
since 2024	**Founder and Director, HealthTech-HL** research group
since 2022	Member of the Commission for Quality Improvement, HSHL
since 2022	**Lead, Human-Machine Interaction research area**, HSHL
since 2022	Member of the Department Council, HSHL
since 2021	**Professor of Computer Science (Immersive Media)**, Hamm-Lippstadt University of Applied Sciences · Director, Immersive Reality Lab
since 2021	Guest Researcher, Technische Universität Berlin
2018–2022	Guest Researcher, DFKI Berlin — artificial intelligence in healthcare
2015–2021	Senior Research Scientist, Technische Universität Berlin
2012	Guest Researcher, INRS Montreal — perception experiments using EEG
2009–2014	Research Scientist, Telekom Innovation Laboratories — subjective and physiological test methods for perceived media quality; six years of research and development outside academia

EDUCATION

2014	**Dr.-Ing.** (Doctor of Engineering), Technische Universität Berlin, Faculty IV — Electrical Engineering and Computer Science, *magna cum laude*. Dissertation: *Neural Correlates of Quality Perception for Complex Speech Signals*, published as a monograph by Springer.
2008	**Diplom-Psychologe** (Psychology), Technische Universität Darmstadt, overall grade *sehr gut*.

AWARDS & RECOGNITION

2025	**Best Paper Award**, MMVE Workshop @ ACM MMSys
2024	**Key Innovator**, Innovation Radar of the European Commission — twice: *Validation Methodology for XR Digital Twin Applications* and *User-Centered Design Methodology for XR Digital Twin Solutions*
2024	Special Prize, Sustainability Award of Volksbank Beckum-Lippstadt (for a supervised Bachelor's thesis)
2023	**Best Journal Paper Award**, IEEE Communications Society (MMTC)
2023	Diversity and Societal Impact Award, QoMEX
2021	Conference Paper Honorable Mention, IEEE VR
2016	Post-doctoral fellowship, Technische Universität Berlin

BIBLIOMETRICS

2,995 citations · h-index 29 · i10-index 83 (Google Scholar, 4 August 2026)

Of these, **2,055 citations** · h-index 23 · i10-index 62 **since 2021** — roughly two thirds of all citations fall in the last five years

1,662 citations from 1,181 documents · 155 indexed publications · h-index 22 (Scopus)

234 publications: 2 monographs and edited volumes · 5 book chapters · 39 peer-reviewed journal articles and 6 further journal contributions · 105 peer-reviewed conference papers and 58 further conference contributions · 17 contributions to international standardization · 2 VDE position papers

STANDARDIZATION & TRANSFER

Co-developer of **ITU-T Recommendation P.812**, *Principles of subjective test methods for interactive virtual reality (VR) applications* (2024)

17 contributions in total: ITU-T Study Group 12 (P.812, P.IntVR, P.PHYSIO, G.1035, P.GAME)

ISO/IEC JTC 1/SC 29/WG 2 (MPEG) — Metaverse use cases

DKE at DIN and VDE — digital qualification in healthcare

2024 founding of XRWise gUG under the BMWK innovation programme for business models and pioneering solutions — a non-profit spin-off carrying XR event tooling out of the research context

THIRD-PARTY FUNDING

Approximately **€4.609 million** in continuous third-party funding since 2015, across **22 funded projects** from 10 funding lines, with awards running through 2029

20 of 22 projects (€4.509 million) applied for and led independently; 3 coordinated as consortium lead

Funders: European Union (Horizon Europe, EIT Digital, Joint Programming Initiative), State of North Rhine-Westphalia (EFRE), BMBF, BMWE/BMWK, BMDV, Innovation Fund of the Federal Joint Committee (G-BA), GKV-Spitzenverband, DAAD, industry partners

Period	Project	Funder	Role	Volume
2026–2029	ITT	EFRE	Subproject lead	€619k
2026–2029	XRT-HuFuSa	EFRE	Consortium coordinator	€289k
2025–2027	Silent Bed Monitor	ZIM	Subproject lead	€218k
2025–2027	XRwise	Innovationsprogramm für Geschäftsmodelle und Pionierlösungen	Subproject lead	€130k
2024–2026	ARiadne	ZIM	Subproject lead	€208k
2024–2025	Fast Jets	Training Optimization Programme	Sole applicant	€44k
2023–2027	Digitalise SÜDWESTFALEN	Innovative Hochschule	Subproject lead	€290k
2023–2025	DIDYMOS-XR	HORIZON Action Grant	Subproject lead	€303k
2023–2025	DigiOnTrack	mFUND	Consortium coordinator	€296k
2023	Body Interaction Lab	HSHL	Sole applicant	€246k
2022–2025	MIA-PROM	KIAS	Subproject lead	€206k
2020	ART	EIT Digital	Subproject lead	€299k
2020	BGF Pflege	Charité / AOK Nordost	Co-applicant	€39k
2019–2020	Mobil im Havelland	Charité / BMBF	Co-applicant	€61k
2019	Physiological workload measurement	Huawei	Sole applicant	€194k
2019	The health score	e-Therapists GmbH	Sole applicant	€20k
2018–2021	DemTab	Innovationsfonds	Subproject lead	€501k
2018–2021	VoiceAdapt	JPI More Years, Better Lives	Consortium coordinator	€256k
2018	GESOBAU workshops	GESOBAU AG	Sole applicant	€9k
2016	Augletics	Augletics GmbH	Sole applicant	€3k

Period	Project	Funder	Role	Volume
2016	Post-doc fellowship	TU Berlin	Sole applicant	€74k
2015–2018	PflegeTab	GKV-Spitzenverband	Subproject lead	€304k
	22 projects			€4,609k

SUPERVISION

5 ongoing doctoral projects in the Immersive Reality Lab, each embedded in a funded project — named topics and completed procedures on the teaching page

4 completed doctoral examinations supervised or reviewed at TU Berlin, including one binational cotutelle with the University of Technology Sydney

35 Master's and 70 Bachelor's theses supervised since 2011

TEACHING

Continuous teaching activity **since 2013**, at Technische Universität Berlin and Hamm-Lippstadt University of Applied Sciences, in German and English

Topics: Augmented Reality · Virtual Reality · Interactive Systems · Graphical User Interfaces · UX Research · Software Ergonomics · Visual Computing · Ubiquitous Computing · Game Development · Introduction to C and C++ · Usability Engineering · Physiological Computing

Module responsibility: *Human Perception* („Mensch und Wahrnehmung“) and *UX Research*

SERVICE & LEADERSHIP

Leadership

Director, Immersive Reality Lab, Hamm-Lippstadt University of Applied Sciences

Mission: developing and evaluating trustworthy XR systems for learning, work and public-space interaction

Team: **11 research staff and 7 student assistants**; annual budget of approximately €1 million

Infrastructure: XR headsets, eye-tracking setups, physiological sensing, behavioural observation and coding

Community & editorial

since 2026	Steering Committee, International Conference on Quality of Multimedia Experience (QoMEX)
2026	Technical Program Chair, MMVE 2026, Hong Kong (co-located with ACM MMSys '26)
2022	General Chair, QoMEX 2022, Lippstadt
since 2025	Editorial Board, *Virtual Worlds* (MDPI)
since 2021	Editorial Board, *Quality and User Experience* (Springer)
2025–2027	Advisory Board, TD-CHAT (Charité Berlin)
2023–2025	Advisory Board, STOP-CSAM (Charité Berlin)

Reviewing & programme committees

Conferences: ACM CHI · IEEE ISMAR · ACM VRST · QoMEX · INTERSPEECH · ICASSP

Journals: IEEE Transactions on Visualization and Computer Graphics · IEEE Transactions on Multimedia · JMIR · Experimental Brain Research · Journal of Building Engineering

Invited talks at industry events, universities and international venues

Joint publications, research proposals and project partnerships with partners in **19 countries**: Australia, Austria, Belgium, Brazil, Canada, Croatia, Denmark, Finland, Greece, Italy, Lebanon, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom, United States

Research stay: INRS Montreal, CA (2012) — Perception experiments using EEG

Named partner institutions

Politecnico di Milano	IT — Social acceptability and interactivity in AR/VR; joint publications and supervision
University of Technology Sydney	AU — Affect recognition in VR, virtual health assistants, digital twins; binational cotutelle doctorate (A. Pinilla, 2022)

American University of Beirut	LB — Horizon Europe project on digital twins for extended reality
NTNU Trondheim	NO — Quality of Experience, immersive media experience, QUALINET
University of Zagreb	HR — ITU-T standardization of interactive VR test methods
University of Toronto	CA — Randomised controlled trial on aphasia training (JMIR)
University of Bremen / St. Gallen	DE/CH — Pseudo-haptics and vibrotactile feedback in VR (ACM CHI 2024)
Queen's University Belfast	UK — Policy work on immersive technologies

SELECTED PUBLICATIONS

Chosen by a fixed rule rather than by preference: every publication carrying an award, then the most recent journal articles. The complete list of 234 — with filters by year, type and topic — is at voigt-antons.de/publications/.

[J33] Da Silva, D. A., Da Silva, A. S., Lima, D. D., Da Costa, J. P., De Melo, L. O., Miranda, C., Santos, G. A., Vinel, A., Mendes, P., Verhoeven, S., Voigt-Antons, J.-N. & De Freitas, E. P. (2026). **Spoofer detection framework for V2X systems via tensor-based DoA estimation and Yolo-based object detection**. IEEE Access.

[J34] Hinzmann, S., Bestgen, A.-K., Schorlemmer, J., Beierlein, C., Kolbe, J. & Voigt-Antons, J.-N. (2026). **A Modular Questionnaire for Target-Group-Specific Evaluation of Event Formats: Developed in the Context of Virtual Worlds Knowledge Transfer**. Virtual Worlds, 5(1), 10.

[J35] Peperkorn, N. L., Ohse, J., Fox, J., Limberg, S., Hadžić, B., Rätsch, M., Voigt-Antons, J.-N., Witthöft, M. & Shiban, Y. (2026). **Effects of Change in Dysfunctional Beliefs and Self-Esteem in Avatar-Based Cognitive Therapy for Symptoms of Social Anxiety Disorder: A Randomized Parallel Trial**. Scientific Reports, 16, 6144.

[J36] Ashrafi, N., Graf, P., Marquardt, M., Harnisch, P., Hillmann, S., Ploner, N., Compagna, D., Cirit, E., Papst, L. & Voigt-Antons, J.-N. (2026). **Multimodal Assistance in Rehabilitation: User Experience of Embodied and Non-Embodied Agents for Collecting Patient-Reported Outcome Measures**. Virtual Worlds, 5(1), 15.

[J37] Haeger, C., Mümken, S. A., Spang, R. P., Brauer, M., O'Sullivan, J. L., Lech, S., Xue, Q.-L., Stockburger, M., Keller, J., Voigt-Antons, J.-N. & Gellert, P. (2026). **Out-of-home mobility enhancement by a physiotherapist-led motivational counseling intervention among rural community-dwelling older adults 75+: The MOBILE RCT**. BMC Geriatrics, 26, 481.

[J38] Henning, J., Hinzmann, S., Vona, F., Amer, M. & Voigt-Antons, J.-N. (2026). **Hearing the Way: Influence of Varying Spatialized Audio Cues and Device Type on AR Navigation Experience**. IEEE Transactions on Visualization and Computer Graphics, pp. 1–8.

[J39] Hallmann, M. C., Peperkorn, N. L., Vona, F., Shiban, Y. & Voigt-Antons, J.-N. (2026). **User Experience and Therapist Effectiveness of Different Virtual Character Types in Virtual Reality Exposure Therapy: A Mixed-Methods Study**. JMIR XR Spatial Comput. — *In press*

[C90] Kojic, T., Vergari, M., Warsinke, M., Ali, D., Möller, S. & Voigt-Antons, J.-N. (2025). **Multimodal User Experience in Extended Reality: Exploring Hand Tracking, Voice, and Passthrough Interactions**. Paper presented at the international workshop on IMmersive Mixed and Virtual Environment Systems (MMVE 2025), Stellenbosch, South Africa. — *Workshop Best Paper Award, MMVE 2025*

[J32] Antoun, M., Elhajj, I. H., Papadopoulos, P. K., Rizk, A., Islam, T., Huerta, I., Toledo, L., Arvanitis, G., Moustakas, K., Jevtić, A., Agustí, I., Vona, F., Voigt-Antons, J.-N., Warsinke, M., Bailer, W., Thallinger, G., Howkins, B., Poder, I., Ruess, M. & Asmar, D. (2025). **Interactive digital twins enabling responsible extended reality applications**. Scientific Reports, 15, 34539.

[OJ6] Kuhlencord, M., Ohse, J., Fox, J., Peperkorn, N., Rätsch, M., Voigt-Antons, J.-N. & Shiban, Y. (2025). **Der Einsatz von vielversprechenden Technologien in der klinisch-psychologischen Diagnostik: Fokus auf KI und XR**. Psychologie in Österreich, (4 & 5), 278-285.

[C71] Ashrafi, N., Schmitt, V., Spang, R. P., Möller, S. & Voigt-Antons, J.-N. (2023). **Protect and Extend - Using GANs for Synthetic Data Generation of Time-Series Medical Records**. Paper presented at the 15th International Conference on Quality of Multimedia Experience (QoMEX 2023), Ghent, Belgium. — *Diversity and Societal Impact Award, QoMEX 2023*

[C66] Vergari, M., Kojić, T., Vona, F., Garzotto, F., Möller, S. & Voigt-Antons, J.-N. (2021). **Influence of Interactivity and Social Environments on User Experience and Social Acceptability in Virtual Reality**. Paper presented at the IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR 2021), Lisbon, Portugal. — *Conference Paper Honorable Mention, IEEE VR 2021*

[J18] Porcu, S., Floris, A., Voigt-Antons, J.-N., Atzori, L. & Möller, S. (2020). **Estimation of the Quality of Experience During Video Streaming From Facial Expression and Gaze Direction**. IEEE Transactions on Network and Service Management, vol. 17, no. 4, pp. 2702-2716. — *Best Journal Paper Award, IEEE Communications Society (MMTC) 2023*

[J6] Engelke, U., Darcy, D. P., Mulliken, G. H., Bosse, S., Martini, M.G. Arndt, S., Antons, J.-N., Chan, K. Y., Ramzan, N. & Brunnström, K. (2017). **Psychophysiology-based QoE Assessment: A Survey**. Journal of Selected Topics in Signal Processing, 11, 6-21. — *Most-cited work*